

Stakeholder Engagement and Mini-Engagement Plan

Medical Advice SLM Challenge

1. Stakeholder Identification

Stakeholder Group 1: Healthcare Professionals and Public-Health Practitioners

This group includes doctors, nurses, pharmacists, public-health practitioners, and other qualified healthcare workers with experience providing health information or making clinical decisions.

They are important because the project is developing an AI system that provides medical and public-health information. Their expertise helps the team determine whether knowledge-base content is medically appropriate, whether model responses are clinically safe, and whether the system correctly distinguishes general health information from situations that require professional care.

They are likely to prioritize patient safety, clinical accuracy, evidence-based information, appropriate escalation, reliable sourcing, medication safety, and clear communication of uncertainty. They will also be concerned that the system does not replace professional judgement or provide misleading diagnosis or treatment recommendations.

The Potential benefits to these group include the Instant, 24/7 access to authoritative guidelines that reduces cognitive load; better tools for patient education; a structured, auditable knowledge base; and support for broader public-health communication and health literacy, while the Potential Risks also include Automation bias and professional deskilling; cultural or contextual mismatch if the model relies solely on Global-North guidelines without accounting for local resource constraints; incorrect medical or medication information causing patient harm; failure to detect emergencies, delaying appropriate care; and unequal information quality from incomplete source coverage.

Stakeholder Group 2: Patients and Community Health Users

This group consists of the people who will ultimately use the Medical Advice SLM to obtain general health and public-health information, including patients, caregivers, families, and members of the wider community for example, a caregiver seeking guidance on infant fever or a pregnant user seeking maternal-health information.

They are important because their experiences reveal whether responses are understandable, culturally appropriate, accessible, and trustworthy. They can surface risks such as algorithmic anxiety, confusing terminology, or culturally insensitive phrasing that may not be visible from a purely technical or clinical perspective.

They are likely to value understandable health information, accessibility, privacy, respect, cultural sensitivity, transparency, and clear guidance about when to seek professional care. They also need confidence that the system will not provide harmful or misleading information.

Ethical, social, and cultural considerations: the system should use respectful, accessible language, avoid unnecessary collection of personal information, acknowledge uncertainty, and avoid assumptions about a user's circumstances. Engagement should include people with different backgrounds and levels of health literacy so the system is not designed only around technically experienced users. Socially, the tool must avoid widening health inequities for example, by defaulting to guidance that assumes resources (transport, facilities, connectivity) that some communities do not have.

PART 2: MINI-ENGAGEMENT PLAN

1. Selected Stakeholder Groups

- Healthcare Professionals and Public-Health Practitioners.
- Patients and Community Health Users.

2. What Matters Most to Them (Stakeholder Interests and Values)

Healthcare professionals care most about clinical accuracy, evidence quality, medication safety, and responsible use of AI. They need confidence that the knowledge base draws on authoritative sources (such as WHO, CDC, and NHS guidance) and that the model does not generate unsupported diagnoses, unsafe medication advice, or inappropriate reassurance. They also value clear escalation procedures for emergencies or situations requiring professional assessment.

Patients and community health users care most about accessibility, understandable information, privacy, cultural relevance, and practical usefulness. They need responses that are easy to understand without losing important safety information, and clear signals about when to seek professional care. Both groups share an interest in a system that is trustworthy, evidence-based, transparent about its limitations, and designed to reduce not introduce health risks.

3. Potential Harms and Potential Benefits

Healthcare Professionals / Public-Health Practitioners

Potential harms: automation bias and professional deskilling; cultural or contextual mismatch if the model relies solely on Global-North guidelines (e.g., NHS-only protocols) without accounting for local resource constraints; incorrect medical or medication information causing patient harm; failure to detect emergencies, delaying appropriate care; and unequal information quality from incomplete source coverage.

Potential benefits: instant, 24/7 access to authoritative WHO/UNICEF/CDC guidelines that reduces cognitive load; better tools for patient education; a structured, auditable knowledge base; and support for broader public-health communication and health literacy.

Patients / Community Health Users

Potential harms: misinterpreting triage cues in ways that delay emergency care; algorithmic anxiety from overly cautious or poorly contextualized outputs; over-reliance on the system in place of professional care; and language that inadvertently stigmatizes certain parenting or health practices, particularly across cultural or linguistic differences.

Potential benefits: greater access to reliable, evidence-based health information; improved health literacy; faster answers to general public-health questions; and better awareness of when to seek in-person medical care, particularly for infant fever, vaccination schedules, and maternal health.

4. Selected Action Catalogue Methods

Methods were selected using the Action Catalogue, a decision-support tool that matches project goals and stakeholder characteristics to appropriate engagement approaches. Based on its guidance, the team selected:

- Healthcare Professionals / Public-Health Practitioners User Committees
- Patients / Community Health Users Participatory Design

5. Why the Chosen Methods Fit Each Stakeholder Group

User Committees for Healthcare Professionals

User Committees are appropriate for healthcare professionals because they possess domain-specific, tacit clinical knowledge that automated evaluation cannot reproduce. The Committee would be a small multidisciplinary group of doctors, nurses, pharmacists, and public-health practitioners who receive a concise description of the project, its intended use, limitations, and safety objectives, then meet on an ongoing basis not just at the end of the project.

During sessions, members would review knowledge-base sources, high-risk questions, medication scenarios, emergency scenarios, pregnancy and infant-health scenarios, and retrieval-supported responses, classifying each as acceptable, requiring revision, requiring escalation, or unsafe using a structured review form (question, response, risk category, identified concern, recommended correction, supporting evidence, final decision). Pharmacists can give particular insight into medication interactions and self-medication risks, while public-health professionals can assess prevention guidance and population-level messaging. This ongoing structure rather than a one-off consultation is what allows the committee to shape the Safety Specification and Source Policy directly, and it builds trust by positioning the SLM as a clinical copilot rather than a top-down imposition on professional practice.

Participatory Design for Patients and Community Health Users

Participatory Design is well suited to patients and community members because health-seeking behavior is emotional and culturally contingent, and a simple usability test cannot capture how different users perceive risk, urgency, or medical authority. The team would recruit a diverse group of potential users and run structured sessions in which participants review example health questions and prototype responses including contrasting versions (e.g., a clinical tone versus an empathetic tone) and discuss whether each response is understandable, complete, respectful, appropriately (not excessively) reassuring, and clear about next steps or the need for professional care.

Participants are not asked to disclose personal medical information; hypothetical or anonymized scenarios are used throughout. Feedback is anonymized, aggregated, and prioritized when multiple participants raise the same concern. This approach surfaces hidden cultural assumptions and communication barriers such as confusing terminology or wording that could unintentionally stigmatize certain practices that neither developers nor clinicians would necessarily catch on their own, ensuring the system is shaped by the people who will actually rely on it.

6. Mini-Engagement Plan Summary

Stakeholder Group	Engagement Method	Main Objective	Activities	Expected Output	Project Stage
Healthcare Professionals / Public-Health Practitioners	User Committees	Validate clinical safety and evidence quality	Review knowledge-base sources, high-risk questions, medication and emergency scenarios, and model-generated outputs	Clinical feedback, identified safety issues, refined evaluation criteria	Dataset development, RAG design, safety evaluation, and final audit
Patients / Community Health Users	Participatory Design	Improve usability, accessibility, and inclusiveness	Review prototype responses; discuss comprehension, tone, and cultural or communication concerns	User feedback, revised response patterns, usability requirements	Dataset refinement, model outputs, and deployment planning

7. Influence on Dataset and Model Design

Feedback from these two engagement methods is intended to directly reshape the project's dataset, model behavior, and evaluation criteria.

- Dataset: healthcare-professional review can surface missing safety-critical topics, incorrect risk classifications, and gaps beyond NHS/WHO-only sourcing, prompting inclusion of more resource-aware, locally relevant protocols. Community feedback can surface underrepresented question types and information needs.
- Model behavior: committee input can shape escalation rules, medication-handling logic, and how the model expresses uncertainty. Community input can shape language complexity, tone, and response structure moving the system's voice from purely clinical toward clear and empathetic.
- Outputs and evaluation: both groups can contribute realistic test cases clinical safety cases from healthcare professionals, and clarity/trust/accessibility cases from community users shifting evaluation beyond automated string-matching toward real-world clinical utility and user trust.
- Deployment: stakeholder feedback can inform which disclaimers are necessary, what human oversight is required, and which use cases (e.g., certain emergency or mental-health scenarios) should be restricted or always escalated to a human.